



Fuglede's Conjecture for a Union of Two Intervals on

$\mathbb{R} \times \mathbb{Z}_N$

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Abstract

In this paper, we prove that a union of two intervals in $\mathbb{R} \times \mathbb{Z}_N$ is a spectral set if and only if it tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations.

Keywords Tiling · spectral sets · Fuglede's conjecture · LCA groups

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1 Introduction

Let G be a locally compact abelian (LCA) group with dual group $\widehat{G}$, and let ν denote the Haar measure on G . A Borel measurable set $\Omega \subset G$ with $0 < \nu(\Omega) < \infty$ is said to be a *spectral set* if there exists a countable set $\Lambda \subset \widehat{G}$ that forms an orthonormal basis for $L^2(\Omega)$. In this case, Λ is called a *spectrum* for Ω , and (Ω, Λ) is referred to as a *spectral pair*.

A measurable set $\Omega \subset G$ is said to *tile* G by translations if there exists a countable set $T \subset G$ such that

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$$\sum_{t \in T} \mathbb{1}_{\Omega}(x - t) = 1 \quad \text{for } \nu \text{ a.e. } x \in G.$$

In this case, T is called a *tiling complement* of Ω , and (Ω, T) is called a *tiling pair*.

Fuglede [15] conjectured that a set $\Omega \subset \mathbb{R}^d$ of finite positive measure is spectral if and only if it tiles $\mathbb{R}^d$ by translations. He proved the equivalence under the additional assumption that the spectrum or the tiling set forms a lattice. However, the general conjecture has been shown to fail in both directions when $d \geq 3$ (see [12, 13, 21, 22, 29, 33]). The conjecture remains open in dimensions $d = 1, 2$, and several partial results have been established. In particular, the conjecture is known to hold for the union of two intervals in one dimension [24], and the “tiling implies spectral” direction has been verified for unions of three intervals [5]. Partial results for the reverse implication have also been obtained [5, 6].

The study of spectral sets and translational tilings fits naturally into the framework of harmonic analysis on locally compact abelian (LCA) groups, where both notions admit canonical formulations in terms of Haar measure, group translations, and characters on the dual group. Substantial progress has been achieved in the setting of finite groups, including cyclic groups $\mathbb{Z}_{p^n}$ [25], vector spaces $\mathbb{Z}_p^d$ (for $p = 2$ and $d \leq 6$; odd primes p and $d = 2$; and $p = 3, 5, 7$ and $d = 3$) [4, 8, 14, 17], as well as product groups such as $\mathbb{Z}_p \times \mathbb{Z}_{p^n}$ [17, 31, 34], $\mathbb{Z}_p \times \mathbb{Z}_{pq}$ [20], $\mathbb{Z}_{pq} \times \mathbb{Z}_{pq}$ [7], $\mathbb{Z}_{p^n q^m}$ (for $p < q$ with $m \leq 9$ or $n \leq 6$, or $p^{m-2} < q^4$) [19, 27, 28], $\mathbb{Z}_{pqr}$ [30], $\mathbb{Z}_{p^2 qr}$ [32], and $\mathbb{Z}_{pqrs}$ [18], where p, q, r, s denote distinct primes.

On the other hand, the concepts of spectral sets and spectral measures have been extended to non-Archimedean locally compact fields, encompassing the field of p -adic numbers and the field of formal Laurent series [9]. The validity of Fuglede’s conjecture for the p -adic field $\mathbb{Q}_p$ has subsequently been verified, and the geometric structure of tiles and spectral sets in $\mathbb{Q}_p$ has further been characterized [10, 11].

The extension of classical density and orthogonality results—for instance, Landau’s necessary density conditions, which in the Euclidean setting yield sharp lower and upper density bounds for sampling and interpolation in Paley–Wiener spaces [26]—to general LCA groups furnishes an important historical and conceptual foundation [16]. This framework also underpins subsequent developments in sampling theory, interpolation, quasicrystals, and Riesz bases of characters over LCA groups; see, for example, [1–3].

In this paper, we study the Fuglede conjecture in the product group $\mathbb{R} \times \mathbb{Z}_N$, where $\mathbb{Z}_N$ denotes the finite cyclic group of order N . Our focus is on the simplest nontrivial geometric configuration in this setting, namely sets formed by two intervals supported on two fibers of $\mathbb{R} \times \mathbb{Z}_N$. This model lies naturally between the classical one-dimensional problem on $\mathbb{R}$ and higher-dimensional or mixed LCA settings: on the one hand, the continuous component still allows the use of essentially one-dimensional methods, including density considerations of Landau type; on the other hand, the discrete component introduces additional arithmetic and combinatorial constraints that do not appear on the real line alone.

This setting is also motivated by related problems beyond additive tilings in product groups. For instance, certain questions on multiplicative tilings of the real line, as studied in [23], can be reformulated as additive tiling problems in $\mathbb{R} \times \mathbb{Z}_2$. More

broadly, the group $\mathbb{R} \times \mathbb{Z}_N$ may be viewed as an intermediate model: it is substantially richer than $\mathbb{R}$, yet remains sufficiently rigid to allow a complete analysis for two-interval sets.

We consider sets $\Omega \subset \mathbb{R} \times \mathbb{Z}_N$ that are unions of two intervals, i.e.

$$\Omega = (a, b) \times \{\ell_1\} \cup (c, d) \times \{\ell_2\},$$

where $(a, b), (c, d) \subset \mathbb{R}$ are intervals and $\ell_1, \ell_2 \in \mathbb{Z}_N$. Our main result verifies the Fuglede conjecture in this setting.

Theorem 1.1 *Let $\Omega \subset \mathbb{R} \times \mathbb{Z}_N$ be the union of two disjoint intervals. Then Ω is a spectral set if and only if it tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations.*

Since the tiling and spectral properties are invariant under translation, reflection, and dilation, we may assume without loss of generality that

$$\Omega = (0, r) \times \{0\} \cup (a, a + 1 - r) \times \{\ell\}, \quad \text{where } \frac{1}{2} \leq r < 1, \quad 0 \leq \ell \leq N - 1. \quad (1.1)$$

When $\ell = 0$, the set Ω is entirely supported on the fiber $\mathbb{R} \times \{0\}$, and the problem reduces to the classical case on the real line, which has already been completely resolved in this setting [24]. Hence, we shall focus on the case $1 \leq \ell \leq N - 1$.

Here we remark that the finite cyclic group $\mathbb{Z}_N$ may be replaced by an arbitrary finite abelian group G . Indeed, the tiling structure depends only on the subgroup generated by ℓ in $\mathbb{Z}_N$ (see Theorem 1.2), which is isomorphic to $\mathbb{Z}_N / \gcd(\ell, N)$, while the “spectrum implies tile” direction follows from the fact that four points on the unit circle with vanishing sum must form a rectangle.

For any integer n , let $v_2(n)$ denote the 2-adic valuation of n , i.e., the largest integer k such that $2^k \mid n$. The following result provides a complete characterization of two-interval tiles in $\mathbb{R} \times \mathbb{Z}_N$.

Theorem 1.2 *Let $\Omega \subset \mathbb{R} \times \mathbb{Z}_N$ be a union of two intervals as in (1.1). Then Ω tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations if and only if one of the following conditions holds:*

- (1) $\frac{1}{2} < r < 1$ and $a - r \in \frac{\gcd(\ell, N)}{N} \mathbb{Z}$;
- (2) $r = \frac{1}{2}$, $a \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z} \setminus \{0\}$, and $v_2(\ell) \geq v_2(N)$;
- (3) $r = \frac{1}{2}$ and $v_2(\ell) < v_2(N)$.

Compared with the case on $\mathbb{R}$, where the geometry of a two-interval set is largely determined by the gap between the intervals, the structure in $\mathbb{R} \times \mathbb{Z}_N$ is more subtle due to the presence of distinct fibers. In particular, the notion of a “gap” is not well-defined across different fiber components. To address this, we carefully analyze whether translated copies of $(0, r) \times \{0\}$ and $(a, a + 1 - r) \times \{\ell\}$ can appear adjacent to each other in a valid tiling configuration.

If Ω satisfies Condition (1), then the translations of the two intervals alternate in the tiling pattern, as shown in Fig. 1. The spectrality of Ω primarily follows from the fact that $((0, 1), \mathbb{Z})$ is a spectral pair.

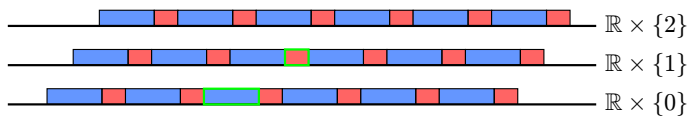


Fig. 1 $\Omega = (0, 7/10) \times \{0\} \cup (1/3 + 7/10, 1/3 + 1) \times \{1\} \subset \mathbb{R} \times \mathbb{Z}_3$

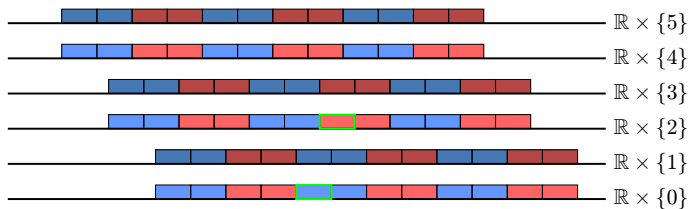


Fig. 2 $\Omega = (0, 1/2) \times \{0\} \cup (1/3, 1/3 + 1/2) \times \{2\} \subset \mathbb{R} \times \mathbb{Z}_6$

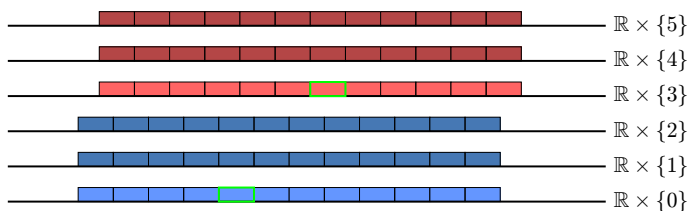


Fig. 3 $\Omega = (0, 1/2) \times \{0\} \cup (a, a + 1/2) \times \{2\} \subset \mathbb{R} \times \mathbb{Z}_6, \forall a \in \mathbb{R}$

If Ω satisfies Condition (2), then the tiling pattern on each line is the same as the tiling pattern of two equal intervals on $\mathbb{R}$, as shown in Fig. 2. The spectrality of Ω primarily follows from the fact that $((0, \frac{1}{2}), 2\mathbb{Z})$ is a spectral pair.

Compared with the case in $\mathbb{R}$, the tiling pattern on each line in Figs. 1 and 2 can be staggered in the vertical direction, rather than being vertically aligned.

Suppose Ω satisfies Condition (3). As shown in Fig. 3, the translations of $(0, r) \times \{0\}$ and $(a, a + 1 - r) \times \{\ell\}$ will not be adjacent. The spectrality and tiling property of Ω primarily follows from the fact that $v_2(\ell) < v_2(N)$ if and only if $\{0, \ell\}$ tiles $\mathbb{Z}_N$ by translations.

Note that the dual group $\widehat{\mathbb{R} \times \mathbb{Z}_N}$ is isomorphic to $\mathbb{R} \times \mathbb{Z}_N$. For $\lambda = (\xi, \eta) \in \mathbb{R} \times \mathbb{Z}_N$, denote

$$e_\lambda(x, y) := e^{2\pi i(\xi x + \frac{\eta}{N} y)},$$

which is a continuous character of the group $\mathbb{R} \times \mathbb{Z}_N$. We identify a point $\lambda \in \mathbb{R} \times \mathbb{Z}_N$ with the point $e_\lambda \in \widehat{\mathbb{R} \times \mathbb{Z}_N}$.

If Λ is a spectrum for Ω , then any translate of Λ is again a spectrum for Ω . In this paper we always assume that

$$(0, 0) \in \Lambda \subset \Lambda - \Lambda.$$

By the orthogonality of the exponential system $\{e_\lambda : \lambda \in \Lambda\}$, we have

$$(0, 0) \in \Lambda \subset \Lambda - \Lambda \subset \mathcal{Z}_\Omega,$$

where $\mathcal{Z}_\Omega$ denotes the zero set of the Fourier transform of the indicator function $\mathbb{1}_\Omega$ together with $(0, 0)$, i.e.,

$$\mathcal{Z}_\Omega = \{(0, 0)\} \cup \{\lambda = (\xi, \eta) \in \mathbb{R} \times \mathbb{Z}_N : \widehat{\mathbb{1}_\Omega}(\xi, \eta) = 0\}.$$

In the proof of the “spectral implies tile” direction, two ingredients play a central role: the geometry of the zero set $\mathcal{Z}_\Omega$, and the Landau-type density theorem for LCA groups due to Gröchenig et al. [16]. The latter provides necessary density conditions for spectra in $\mathbb{R} \times \mathbb{Z}_N$, and when combined with the structural properties of $\mathcal{Z}_\Omega$, it implies that the existence of a spectrum forces Ω to satisfy one of the conditions in Theorem 1.2.

2 The Structure of Tiles for Two Intervals

In this section we assume Ω satisfies (1.1). Recall that

$$\Omega = (0, r) \times \{0\} \cup (a, a + 1 - r) \times \{\ell\}, \quad \frac{1}{2} \leq r < 1, \quad 1 \leq \ell \leq N - 1.$$

Denote $I_0 = (0, r) \times \{0\}$, $I_1 = (a, a + 1 - r) \times \{\ell\}$.

Suppose Ω tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations with tiling set T . Without loss of generality, we assume $(0, 0) \in T$. Denote

$$\mathcal{T} = \{I_i + t : i = 0, 1, t \in T\}.$$

We discuss the structure of tiles in two cases: $\frac{1}{2} < r < 1$ and $r = 1/2$.

2.1 Case 1: $\frac{1}{2} < r < 1$

For $\frac{1}{2} < r < 1$, the following lemma provides the structure of tiles.

Lemma 2.1 *Suppose Ω tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations and $\frac{1}{2} < r < 1$. Then the translates of I_0 and I_1 must alternate in $\mathcal{T}$.*

Proof If $\mathcal{T}$ contains two consecutive translates $I_1 + t$ and $I_1 + t + (1 - r, 0)$ of I_1 , it would also contain the matching translates $I_0 + t$ and $I_0 + t + (1 - r, 0)$ of I_0 , which is impossible since the latter two intervals overlap.

Now suppose that $\mathcal{T}$ contains two consecutive translates $I_0 + t$ and $I_0 + t + (r, 0)$ of I_0 , then $\mathcal{T}$ must contain the matching translates $I'_1 = I_1 + t$ and $I''_1 = I_1 + t + (r, 0)$ of I_1 . The gap between I'_1 and I''_1 has length $2r - 1$, which is strictly less than r , so that I'_1 must be followed by another translate of I_1 . However, it follows from the above that this is impossible. $\square$

Now by Lemma 2.1, we give the relationship between a and r when Ω tiles $\mathbb{R} \times \mathbb{Z}_N$.

Lemma 2.2 Assume that $\frac{1}{2} < r < 1$. If Ω tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations, then

$$a - r \in \frac{\gcd(\ell, N)}{N} \mathbb{Z}.$$

Moreover, the tiling set T has the form of

$$T = \bigcup_{j=0}^{N-1} (\mathbb{Z} + r_j) \times \{j\}$$

for some $r_j \in \mathbb{R}$.

Proof By Lemma 2.1, we know that when $\frac{1}{2} < r < 1$ the translations of I_0 and I_1 must alternate. Because the sum of the length of I_0 and I_1 is 1, we conclude that

$$T = \bigcup_{j=0}^{N-1} (\mathbb{Z} + r_j) \times \{j\},$$

where $r_0 = 0$ by the assumption that $(0, 0) \in T$.

Since $(r_j, j) \in T$ for $0 \leq j \leq N-1$, there is a translation of I_0 given by $I_0 + (r_j, j)$. By Lemma 2.1, $I_0 + (r_j, j)$ is followed by the translation $I_1 + (r + r_j - a, j - \ell)$ of I_1 , which implies $(r_j + r - a, j - \ell) \in T$. Then

$$r_j + r - a \equiv r_{j-\ell} \pmod{1}.$$

By successively taking $j = \ell, 2\ell, \dots, \frac{N\ell}{\gcd(\ell, N)}$, we obtain

$$\begin{cases} r_\ell \equiv r_0 + a - r \pmod{1}, \\ r_{2\ell} \equiv r_\ell + a - r \pmod{1}, \\ \vdots \\ r_{\frac{N\ell}{\gcd(\ell, N)}} \equiv r_{\left(\frac{N}{\gcd(\ell, N)} - 1\right)\ell} + a - r \pmod{1}. \end{cases} \quad (2.1)$$

Since $\frac{N\ell}{\gcd(\ell, N)} \equiv 0 \pmod{N}$, summing equations in (2.1), we get $\frac{N(a-r)}{\gcd(\ell, N)} \equiv 0 \pmod{1}$. Then

$$a - r \in \frac{\gcd(\ell, N)}{N} \mathbb{Z}.$$

Moreover, the tiling set T can be chosen as

$$T = \bigcup_{i=0}^{\gcd(\ell, N)-1} \bigcup_{j=0}^{\frac{N}{\gcd(\ell, N)}-1} (\mathbb{Z} + j(a - r)) \times \{j\ell + i\}.$$

□

2.2 Case 2: $r = \frac{1}{2}$

Recall that $v_2(n)$ denote the 2-adic valuation of an integer n . Note that $v_2(\ell) < v_2(N)$ if and only if $\{0, \ell\}$ tiles $\mathbb{Z}_N$ by translations. For this reason, we discuss the structure of tiles when $r = \frac{1}{2}$ in two cases: $v_2(\ell) < v_2(N)$ and $v_2(\ell) \geq v_2(N)$.

Lemma 2.3 *Assume that $r = \frac{1}{2}$ and $v_2(\ell) < v_2(N)$. Then Ω tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations.*

Proof It suffices to take

$$T = \frac{1}{2}\mathbb{Z} \times \left\{ 2j\ell + i : 0 \leq j < \frac{N}{2\gcd(\ell, N)}, 0 \leq i < \gcd(\ell, N) \right\}.$$

□

Now suppose $v_2(\ell) \geq v_2(N)$. The following lemma provides the condition that a must satisfy when Ω tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations.

Lemma 2.4 *Suppose that $r = \frac{1}{2}$ and $v_2(\ell) \geq v_2(N)$. If Ω tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations, then*

$$a \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z} \setminus \{0\}.$$

Proof Suppose $\mathbb{R} \times \{0\}$ is completely covered by translations of I_0 . Then $\frac{1}{2}\mathbb{Z} \times \{0\} \subset T$, and consequently, $\mathbb{R} \times \{\ell\}$ is completely covered by translations of I_1 . Then $\mathbb{R} \times \{2\ell\}$ cannot contain a translation of I_1 , as this would imply a corresponding translation of I_0 on $\mathbb{R} \times \{\ell\}$, which is impossible. By induction, $\mathbb{R} \times \{2j\ell\}$ is completely covered by translations of I_0 , while $\mathbb{R} \times \{(2j+1)\ell\}$ is completely covered by translations of I_1 .

Since $\frac{N}{\gcd(\ell, N)}$ is an odd number, we can choose $j = \frac{1}{2} \left(\frac{N}{\gcd(\ell, N)} - 1 \right)$ such that $2j\ell = \left(\frac{N}{\gcd(\ell, N)} - 1 \right)\ell$ and thus $\mathbb{R} \times \left\{ \left(\frac{N}{\gcd(\ell, N)} - 1 \right)\ell \right\}$ is completely covered by translations of I_0 . Consequently, since $\frac{N}{\gcd(\ell, N)}\ell \equiv 0 \pmod{N}$, it follows that $\mathbb{R} \times \{0\}$ would be completely covered by translations of I_1 , a contradiction.

Then no single line can be completely covered by translations of I_0 , which implies that every line contains consecutive translates of I_0 and I_1 . We can choose $(t_{j\ell}, j\ell) \in T$ such that $I_0 + (t_{j\ell}, j\ell)$, $0 \leq j < \frac{N}{\gcd(\ell, N)}$ are translates of I_0 . Clearly, $I_1 + (t_{j\ell}, j\ell)$ is a translation of I_1 . Then the gap between $I_1 + (t_{j\ell}, j\ell)$ and $I_0 + (t_{(j+1)\ell}, (j+1)\ell)$ must then be covered by non-overlapping intervals of length $\frac{1}{2}$. This results in the

system of congruences:

$$\begin{cases} a + t_0 \equiv t_\ell \left(\bmod \frac{1}{2} \right), \\ a + t_\ell \equiv t_{2\ell} \left(\bmod \frac{1}{2} \right), \\ \vdots \\ a + t \left(\frac{N}{\gcd(\ell, N)} - 1 \right) \ell \equiv t \frac{N\ell}{\gcd(\ell, N)} \left(\bmod \frac{1}{2} \right). \end{cases} \quad (2.2)$$

Summing up equations in (2.2) yields: $\frac{Na}{\gcd(\ell, N)} \equiv 0 \left(\bmod \frac{1}{2} \right)$, which implies

$$a \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z}.$$

If $a = 0$, then $\Omega = \left(0, \frac{1}{2} \right) \times \{0, \ell\}$. Since $I_1 \in \mathcal{T}$, $\left(0, \frac{1}{2} \right) \times \{2\ell\}$ can only be covered by translates of I_0 and $\left(0, \frac{1}{2} \right) \times \{3\ell\}$ should be covered by translates of I_1 . Since $\frac{N}{\gcd(\ell, N)}$ is an odd integer, by induction, $\left(0, \frac{1}{2} \right) \times \left\{ \left(\frac{N}{\gcd(\ell, N)} - 1 \right) \ell \right\}$ must be covered by translates of I_0 and $I_0 = \left(0, \frac{1}{2} \right) \times \left\{ \frac{N}{\gcd(\ell, N)} \ell \right\}$ must be covered by copies of I_1 , contradicting the fact that $I_0 \in \mathcal{T}$. Therefore, Ω cannot tile $\mathbb{R} \times \mathbb{Z}_N$ by translations. Thus

$$a \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z} \setminus \{0\}.$$

Suppose $a = \frac{\gcd(\ell, N)}{2N} n$ for some $n \in \mathbb{Z} \setminus \{0\}$. Write $n = 2^t s$, $\gcd(2, s) = 1$. Then, we can choose T as

$$T = \bigcup_{i=0}^{\gcd(\ell, N)-1} \bigcup_{j=0}^{\frac{N}{\gcd(\ell, N)}-1} \bigcup_{k=0}^{2^t-1} \left(2^t (\mathbb{Z} + j(a-r)) + \frac{k}{2} \right) \times \{j\ell + i\}.$$

□

Combining Lemmas 2.2, 2.3 and 2.4, we obtain Theorem 1.2.

3 Tiling Implies Spectral

Let $\mathbb{R} \times \mathbb{Z}_N$ be the product group of the real line $\mathbb{R}$ and cyclic group $\mathbb{Z}_N = \{0, \dots, N-1\}$. It is a locally compact abelian group. The Haar measure $\nu = \mathfrak{m} \times \mu$ on $\mathbb{R} \times \mathbb{Z}_N$

is the product measure of the Lebesgue measure m on $\mathbb{R}$ and the counting measure μ on $\mathbb{Z}_N$.

For $\lambda = (\xi, \eta) \in \mathbb{R} \times \mathbb{Z}_N$, denote

$$e_\lambda(x, y) = e^{2\pi i \left(\xi x + \frac{\eta}{N} y \right)},$$

which is a continuous character of the group $\mathbb{R} \times \mathbb{Z}_N$. For $f \in L^1(\mathbb{R} \times \mathbb{Z}_N)$, its Fourier transform $\widehat{f}$ is defined as

$$\widehat{f}(\lambda) = \int_{\mathbb{R} \times \mathbb{Z}_N} f(x, y) e_{-\lambda}(x, y) dv.$$

For a measurable set $\Omega \subset \mathbb{R} \times \mathbb{Z}_N$, denote by

$$\mathcal{Z}_\Omega = \{(0, 0)\} \cup \{\lambda = (\xi, \eta) \in \mathbb{R} \times \mathbb{Z}_N : \widehat{\mathbb{1}_\Omega}(\xi, \eta) = 0\}$$

the set of zeros of the Fourier transform of the indicator functions $\mathbb{1}_\Omega$ together with $(0, 0)$.

Now suppose Ω satisfies (1.1), that is

$$\Omega = (0, r) \times \{0\} \cup (a, a + 1 - r) \times \{\ell\}, \quad \frac{1}{2} \leq r < 1, \quad 1 \leq \ell \leq N - 1.$$

In this section, we show the spectrality of the tiles for the union of two intervals.

3.1 Zeros of the Fourier Transform of the Indicator Function

To verify the orthogonality of the spectrum, we need to calculate the set $\mathcal{Z}_\Omega$. Let

$$Z_0 = \{(0, j) : 0 < j < N\}.$$

We begin by examining the relationship between $\mathcal{Z}_\Omega$ and Z_0 . The following lemma provides a necessary condition involving the parameters r , N , and ℓ when $\mathcal{Z}_\Omega \cap Z_0 \neq \emptyset$.

Lemma 3.1 *If $\mathcal{Z}_\Omega \cap Z_0 \neq \emptyset$, then $r = \frac{1}{2}$, and $v_2(\ell) < v_2(N)$. Moreover, we have*

$$2\mathbb{Z} \times \left(\{0\} \cup \left\{ \eta \in \mathbb{Z}_N : \ell\eta \equiv \frac{N}{2} \pmod{N} \right\} \right) \subset \mathcal{Z}_\Omega.$$

Proof If $(0, \eta) \in \mathcal{Z}_\Omega \cap Z_0$, then

$$0 = \widehat{\mathbb{1}_\Omega}((0, \eta)) = \int_0^r 1 dx + e^{-2\pi i \frac{\eta\ell}{N}} \int_a^{a+1-r} 1 dx = r + e^{-2\pi i \frac{\eta\ell}{N}} (1 - r).$$

It follows that

$$r = \frac{1}{1 - e^{2\pi i \cdot \frac{\eta \ell}{N}}} \in \left[\frac{1}{2}, 1 \right).$$

Thus $e^{2\pi i \frac{\eta \ell}{N}} = -1$, which implies that $\frac{\eta \ell}{N} \in \mathbb{Z} + \frac{1}{2}$ and $r = \frac{1}{2}$.

Consider the congruence equation

$$\ell \eta \equiv \frac{N}{2} \pmod{N}. \quad (3.1)$$

It has a solution, implying that $\gcd(\ell, N) \mid \frac{N}{2}$, which is equivalent to $v_2(\ell) < v_2(N)$. $\square$

Now we consider the case $\mathcal{Z}_\Omega \cap Z_0 = \emptyset$. The following lemma provides the zero set of the Fourier transform of the indicator function of Ω .

Lemma 3.2 *If $\mathcal{Z}_\Omega \cap Z_0 = \emptyset$, then $\mathcal{Z}_\Omega = (Z_1 \cup Z_2 \cup Z_3) \setminus Z_0$, where*

$$\begin{aligned} Z_1 &= \bigcup_{j=0}^{N-1} \left\{ \xi \in \mathbb{R} : \xi a \in \mathbb{Z} + \frac{1}{2} - \frac{j\ell}{N}, \xi(1-2r) \in \mathbb{Z} \right\} \times \{j\}, \\ Z_2 &= \{\xi \in \mathbb{Z} : \xi r \in \mathbb{Z}\} \times \mathbb{Z}_N, \\ Z_3 &= \bigcup_{j=0}^{N-1} \left\{ \xi \in \mathbb{Z} : \xi(a-r) \in \mathbb{Z} - \frac{j\ell}{N} \right\} \times \{j\}. \end{aligned}$$

Proof Suppose that $\lambda = (\xi, \eta) \in \mathcal{Z}_\Omega$ with $\lambda \neq (0, 0)$. Then

$$\widehat{\mathbb{1}}_\Omega(\xi, \eta) = \int_\Omega e^{-2\pi i(\xi x + \frac{\eta}{N} y)} d(m \times \mu) = \int_0^r e^{-2\pi i \xi x} dx + e^{-2\pi i \frac{\eta \ell}{N}} \int_a^{a+1-r} e^{-2\pi i \xi x} dx = 0.$$

Since $\mathcal{Z}_\Omega \cap Z_0 = \emptyset$, then $\xi \neq 0$, and

$$\widehat{\mathbb{1}}_\Omega(\xi, \eta) = -\frac{1}{2\pi i \xi} \left(e^{-2\pi i \xi r} - 1 + e^{-2\pi i \left(\frac{\eta \ell}{N} + \xi(a+1-r) \right)} - e^{-2\pi i \left(\frac{\eta \ell}{N} + \xi a \right)} \right) = 0.$$

It is well known that if four points lying on the unit circle sum to 0, then they form a rectangle. The solutions to $z_1 + z_2 + z_3 + 1 = 0$, with $|z_i| = 1$, must be of the form $\{z_1, z_2, z_3\} = \{-1, z, -z\}$ for some z on the unit circle. Hence, $\lambda \in \mathcal{Z}_\Omega$ if one of the following holds:

- (1) $e^{2\pi i \left(\frac{\eta \ell}{N} + \xi a \right)} = -1, e^{2\pi i \xi r} = -e^{2\pi i \left(\frac{\eta \ell}{N} + \xi(a+1-r) \right)}$, which implies $\lambda \in Z_1$;
- (2) $e^{2\pi i \xi r} = 1, e^{2\pi i \left(\frac{\eta \ell}{N} + \xi(a+1-r) \right)} = e^{2\pi i \left(\frac{\eta \ell}{N} + \xi a \right)}$, which implies $\lambda \in Z_2$;

$$(3) \quad e^{2\pi i \xi r} = e^{2\pi i \left(\frac{\eta \ell}{N} + \xi a \right)}, e^{2\pi i \left(\frac{\eta \ell}{N} + \xi(a+1-r) \right)} = 1, \text{ which implies } \lambda \in Z_3.$$

□

Clearly, Z_2 and Z_3 are subgroups of $\mathbb{Z} \times \mathbb{Z}_N$. And since the intersection of two subgroups is still a subgroup, $Z_2 \cap Z_3$ is a subgroup of $\mathbb{Z} \times \mathbb{Z}_N$.

3.2 Tiling Implies Spectral

Note that in the case $a - r \in \mathbb{Z}$, one can verify that $\Lambda = \mathbb{Z} \times \{0\}$ is a spectrum for Ω , as in the real line.

Assume that Ω tiles $\mathbb{R} \times \mathbb{Z}_N$ by translations. By Theorem 1.2, we distinguish between three cases:

- (1) $1/2 < r < 1$ and $a - r \in \frac{\gcd(\ell, N)}{N} \mathbb{Z} \setminus \{0\}$;
- (2) $r = \frac{1}{2}$, $a \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z} \setminus \{0\}$ and $v_2(\ell) \geq v_2(N)$;
- (3) $r = \frac{1}{2}$ and $v_2(\ell) < v_2(N)$.

3.2.1 $1/2 < r < 1$ and $a - r \in \frac{\gcd(\ell, N)}{N} \mathbb{Z}$

For this case, we choose the spectrum Λ of Ω from Z_3 . Recall that

$$Z_3 = \bigcup_{j=0}^{N-1} \left\{ \xi \in \mathbb{Z} : \xi(a - r) \in \mathbb{Z} - \frac{j\ell}{N} \right\} \times \{j\}.$$

Note that when $a - r \notin \mathbb{Q}$, Z_3 contains only finitely many points and cannot form a spectrum. To select the spectrum Λ , we calculate Z_3 under the assumption that $a - r \in \mathbb{Q}$ in the following lemma, which will also be used in the proof of “spectral implies tiling.”

Lemma 3.3 *Suppose Ω satisfies*

$$a - r = \frac{q}{p} \in \mathbb{Q} \setminus \{0\}, \quad \text{with } p > 0, \quad \gcd(p, q) = 1.$$

Then

$$Z_3 = \bigcup_{j=0}^{\gcd(p\ell, N)-1} \left(p\mathbb{Z} + jq_* \cdot \frac{p\ell}{\gcd(p\ell, N)} \right) \times \left\{ \frac{jN}{\gcd(p\ell, N)} \right\},$$

where $q_* = \frac{p^{\phi(q)} - 1}{q} \in \mathbb{Z}$.

Proof Suppose $Z_3 \cap (\mathbb{Z} \times \{j\}) \neq \emptyset$. By the definition of Z_3 , this implies that there exists $\xi \in \mathbb{R}$ such that

$$\xi \in (a - r)^{-1} \left(\mathbb{Z} - \frac{j\ell}{N} \right) \cap \mathbb{Z}.$$

Since $a - r = \frac{q}{p}$, the condition above is equivalent to

$$\xi \in \frac{\left(p\mathbb{Z} - \frac{jp\ell}{N}\right) \cap q\mathbb{Z}}{q}, \quad (3.2)$$

which means that the congruence equation

$$pn \equiv \frac{jp\ell}{N} \pmod{|q|} \quad (3.3)$$

has a solution. It follows that $\frac{jp\ell}{N} \in \mathbb{Z}$, or equivalently,

$$j \in \frac{N}{\gcd(p\ell, N)}\mathbb{Z}. \quad (3.4)$$

There are $\gcd(p\ell, N)$ distinct values of j satisfy (3.4) in the sense of modulo N .

The solution to the congruence equation (3.3) is given by

$$n \equiv p^{\phi(|q|)-1} \cdot \frac{jp\ell}{N} \pmod{|q|}. \quad (3.5)$$

Thus, when n satisfies (3.5), we have $pn - \frac{jp\ell}{N} \in q\mathbb{Z}$. Substituting (3.5) into (3.2), we obtain

$$\xi \in \frac{p\left(p^{\phi(|q|)-1} \cdot \frac{jp\ell}{N} + q\mathbb{Z}\right) - \frac{jp\ell}{N}}{q} = p\mathbb{Z} + \frac{jp\ell}{N} \cdot \frac{p^{\phi(|q|)} - 1}{q}. \quad (3.6)$$

We remark that by Euler's theorem, $\frac{p^{\phi(|q|)} - 1}{q} \in \mathbb{Z}$.

In summary, combining (3.4) and (3.6), we have

$$Z_3 = \bigcup_{j=0}^{\gcd(p\ell, N)-1} \left(p\mathbb{Z} + j \cdot \frac{p^{\phi(|q|)} - 1}{q} \cdot \frac{p\ell}{\gcd(p\ell, N)} \right) \times \left\{ \frac{jN}{\gcd(p\ell, N)} \right\}.$$

□

Based on the specific expression of the set Z_3 , we select the spectrum Λ from Z_3 for Ω .

Proposition 3.4 *Suppose Ω satisfies*

$$a - r = \frac{q}{p} \in \frac{\gcd(\ell, N)}{N}\mathbb{Z} \setminus \{0\}, \quad \text{with } p > 0, \quad \gcd(p, q) = 1.$$

Then Ω is a spectral set with spectrum

$$\Lambda = \bigcup_{j=0}^{p-1} \left(p\mathbb{Z} + jq_* \cdot \frac{\ell}{\gcd(\ell, N)} \right) \times \left\{ \frac{jN}{p \gcd(\ell, N)} \right\},$$

where $q_* = \frac{p^{\phi(lq)} - 1}{q} \in \mathbb{Z}$.

Proof Since $p \mid \frac{N}{\gcd(\ell, N)}$, we have $\gcd(p\ell, N) = p \gcd(\ell, N)$. By Lemma 3.3, we have $\Lambda \subset Z_3$. Noting that Z_3 is a subgroup of $\mathbb{Z} \times \mathbb{Z}_N$, we conclude that $\Lambda - \Lambda \subset Z_3$.

Since $p \mid \frac{N}{\gcd(\ell, N)}$ and $\gcd\left(\frac{N}{\gcd(\ell, N)}, \frac{\ell}{\gcd(\ell, N)}\right) = 1$, it follows that $\gcd\left(p, \frac{\ell}{\gcd(\ell, N)}\right) = 1$. Using the fact that $\gcd(p^{\phi(lq)} - 1, p) = 1$, we have $\gcd(q_*, p) = 1$. Therefore, the set

$$\left\{ jq_* \cdot \frac{\ell}{\gcd(\ell, N)} : 0 \leq j < p \right\}$$

forms a complete residue system modulo p . Then the elements in this set are pairwise distinct in the sense of modulo p . Therefore,

$$\Lambda - \Lambda \subset Z_3 \setminus Z_0,$$

and the orthogonality of Λ follows from Lemma 3.2.

Next, we check the completeness of Λ , that is, every $f \in L^2(\Omega)$ can be uniquely represented as:

$$f(x, y) = \sum_{\lambda \in \Lambda} a_\lambda e_\lambda(x, y),$$

with $a_\lambda = \int_\Omega f(x, y) e_{-\lambda}(x, y) \, dv$.

For any $f \in L^2(\Omega)$, let

$$h(x) = \begin{cases} f(x, 0), & x \in (0, r), \\ f(x + a - r, \ell), & x \in (r, 1). \end{cases}$$

Then $h \in L^2((0, 1))$. Since $((0, 1), \mathbb{Z})$ forms a spectral pair, $h(x)$ can be expanded as

$$h(x) = \sum_{n \in \mathbb{Z}} c_n e^{2\pi i \cdot nx}, \quad x \in (0, 1),$$

where

$$c_n = \int_0^1 h(x) e^{-2\pi i \cdot nx} \, dx.$$

Noting that the set

$$\left\{ jq_* \cdot \frac{\ell}{\gcd(\ell, N)} : 0 \leq j < p \right\}$$

forms a complete residue system modulo p , we can partition $\mathbb{Z}$ into p equivalence classes of $\mathbb{Z}_p$:

$$\mathbb{Z} = \bigcup_{j=0}^{p-1} \bigcup_{k \in \mathbb{Z}} \left(pk + jq_* \cdot \frac{\ell}{\gcd(\ell, N)} \right).$$

For $0 \leq j < p$ and $k \in \mathbb{Z}$, denote

$$\xi_{k,j} = pk + jq_* \cdot \frac{\ell}{\gcd(\ell, N)}$$

and

$$\lambda_{k,j} = \left(\xi_{k,j}, \frac{jN}{p \gcd(\ell, N)} \right).$$

It follows that

$$h(x) = \sum_{n \in \mathbb{Z}} c_n e^{2\pi i \cdot nx} = \sum_{j=0}^{p-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x, 0), \quad x \in (0, 1),$$

where $a_{\lambda_{k,j}} = c_{\xi_{k,j}}$. Then, we have

$$f(x, 0) = h(x) = \sum_{j=0}^{p-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x, 0), \quad x \in (0, r), \quad (3.7)$$

and

$$f(x, \ell) = h(x - a + r) = \sum_{j=0}^{p-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x - a + r, 0), \quad x \in (a, a + 1 - r).$$

Since

$$\begin{aligned} \xi_{k,j}(r - a) &\equiv \left(pk + j \cdot \frac{p^{\phi(|q|)} - 1}{q} \cdot \frac{\ell}{\gcd(\ell, N)} \right) \cdot \left(-\frac{q}{p} \right) \\ &\equiv -qk - j \cdot p^{\phi(|q|)-1} \cdot \frac{\ell}{\gcd(\ell, N)} + \frac{j\ell}{p \gcd(\ell, N)} \\ &\equiv \frac{j\ell}{p \gcd(\ell, N)} \pmod{1}, \end{aligned}$$

it follows that

$$e_{\lambda_{k,j}}(x + r - a, 0) = e_{\lambda_{k,j}}(x, \ell), \quad x \in (a, a + 1 - r).$$

Therefore, we have

$$f(x, \ell) = \sum_{j=0}^{p-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x, \ell), \quad x \in (a, a+1-r). \quad (3.8)$$

Combining (3.7) and (3.8), we obtain

$$f(x, y) = \sum_{j=0}^{p-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x, y), \quad (x, y) \in \Omega.$$

□

3.2.2 $r = \frac{1}{2}$, $a \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z} \setminus \{0\}$ and $\nu_2(\ell) \geq \nu_2(N)$

To select the spectrum Λ , we calculate Z_1 and $Z_2 \cap Z_3$ under the assumption that $r = \frac{1}{2}$ and $a \in \mathbb{Q} \setminus \{0\}$ in the following lemma,

Lemma 3.5 *Suppose Ω satisfies*

$$r = \frac{1}{2}, \quad a = \frac{q}{p} \in \mathbb{Q} \setminus \{0\}, \quad \text{with } p > 0, \quad \gcd(p, q) = 1.$$

Then

$$Z_2 \cap Z_3 = \bigcup_{j=0}^{\gcd\left(\frac{p}{d} \cdot \ell, N\right)-1} 2 \left(\frac{p}{d} \mathbb{Z} + jq_* \cdot \frac{\frac{p}{d} \cdot \ell}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \right) \times \left\{ \frac{jN}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \right\},$$

where $d = \gcd(p, 2)$ and $q_* = \frac{\left(\frac{p}{d}\right)^{\phi\left(\frac{2|q|}{d}\right)} - 1}{\frac{2q}{d}} \in \mathbb{Z}$.

Moreover,

$$(Z_2 \cap Z_3 + (\xi_*, \eta_*)) \subset Z_1,$$

where $(\xi_*, \eta_*) \in \mathbb{R} \times \mathbb{Z}_N$ satisfies $\xi_* a + \frac{\eta_* \ell}{N} \in \mathbb{Z} + \frac{1}{2}$.

Proof By the definition of Z_2 and Z_3 , we have

$$Z_2 \cap Z_3 = \bigcup_{j=0}^{N-1} \left\{ \xi \in 2\mathbb{Z} : \xi a \in \mathbb{Z} - \frac{j\ell}{N} \right\} \times \{j\}.$$

Assume that $(Z_2 \cap Z_3) \cap (2\mathbb{Z} \times \{j\}) \neq \emptyset$, which implies that there exists $\xi \in \mathbb{R}$ such that

$$\xi \in 2\mathbb{Z} \cap a^{-1} \left(\mathbb{Z} - \frac{j\ell}{N} \right).$$

Since $a = \frac{q}{p}$, the condition above is equivalent to

$$\xi \in \frac{2q\mathbb{Z} \cap \left(p\mathbb{Z} - \frac{jp\ell}{N}\right)}{q}. \quad (3.9)$$

Consequently, the congruence equation

$$pn \equiv \frac{jp\ell}{N} \pmod{2|q|} \quad (3.10)$$

has a solution, which implies that

$$\gcd(p, 2q) \mid \frac{jp\ell}{N}. \quad (3.11)$$

Since $\gcd(p, q) = 1$, we have $\gcd(p, 2q) = \gcd(p, 2)$. Recall that $d = \gcd(p, 2)$. Then by (3.11), we have $\frac{jp\ell}{dN} \in \mathbb{Z}$, which is equivalent to

$$j \in \frac{N}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \mathbb{Z}. \quad (3.12)$$

And there are $\gcd\left(\frac{p}{d} \cdot \ell, N\right)$ distinct j satisfy (3.12) in the sense of modulo N .

The solution to the congruence equation (3.10) is given by

$$n \equiv \left(\frac{p}{d}\right)^{\phi\left(\frac{2|q|}{d}\right)-1} \cdot \frac{jp\ell}{dN} + \frac{2|q|}{d}t \pmod{2|q|}, \quad 0 \leq t \leq d-1. \quad (3.13)$$

Thus, when n satisfies (3.13), $p\mathbb{Z} - \frac{jp\ell}{N} \in 2q\mathbb{Z}$. Substituting (3.13) into (3.9), we obtain

$$\xi \in \frac{p \left(\left(\frac{p}{d}\right)^{\phi\left(\frac{2|q|}{d}\right)-1} \cdot \frac{jp\ell}{dN} + \frac{2q}{d}\mathbb{Z}\right) - \frac{jp\ell}{N}}{q} = 2 \left(\frac{p}{d}\mathbb{Z} + \frac{jp\ell}{dN} \cdot \frac{\left(\frac{p}{d}\right)^{\phi\left(\frac{2|q|}{d}\right)} - 1}{\frac{2q}{d}} \right). \quad (3.14)$$

Note that by Euler's theorem, $\frac{\left(\frac{p}{d}\right)^{\phi\left(\frac{2|q|}{d}\right)} - 1}{\frac{2q}{d}} \in \mathbb{Z}$.

Combining (3.12) and (3.14), we have

$$Z_2 \cap Z_3 = \bigcup_{j=0}^{\gcd\left(\frac{p}{d} \cdot \ell, N\right)-1} 2 \left(\frac{p}{d}\mathbb{Z} + j \cdot \frac{\left(\frac{p}{d}\right)^{\phi\left(\frac{2|q|}{d}\right)} - 1}{\frac{2q}{d}} \cdot \frac{\frac{p}{d} \cdot \ell}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \right) \times \left\{ \frac{jN}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \right\}.$$

Since $r = \frac{1}{2}$, by Lemma 3.2,

$$Z_1 = \bigcup_{j=0}^{N-1} \left\{ \xi \in \mathbb{R} : \xi a \in \mathbb{Z} + \frac{1}{2} - \frac{j\ell}{N} \right\}.$$

For any $(\xi, \eta) \in Z_2 \cap Z_3$, we have

$$\xi a \in \mathbb{Z} - \frac{\eta\ell}{N}.$$

Since

$$\xi_* a \in \mathbb{Z} + \frac{1}{2} - \frac{\eta_*\ell}{N},$$

it follows that

$$(\xi + \xi_*)a \in \mathbb{Z} + \frac{1}{2} - \frac{(\eta + \eta_*)\ell}{N}.$$

Therefore, $(\xi + \xi_*, \eta + \eta_*) \in Z_1$. And then

$$Z_2 \cap Z_3 + (\xi_*, \eta_*) \subset Z_1.$$

□

Proposition 3.6 Suppose Ω satisfies

$$r = \frac{1}{2}, a = \frac{q}{p} \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z} \setminus \{0\}, \quad \text{with } p > 0, \quad \gcd(p, q) = 1,$$

and $v_2(\ell) \geq v_2(N)$. Then Ω is a spectral set with spectrum

$$\begin{aligned} \Lambda = & \bigcup_{j=0}^{\frac{p}{d}-1} 2 \left(\frac{p}{d} \mathbb{Z} + jq_* \cdot \frac{\ell}{\gcd(\ell, N)} \right) \times \left\{ \frac{jdN}{p \gcd(\ell, N)} \right\} \\ & \cup \bigcup_{j=0}^{\frac{p}{d}-1} 2 \left(\frac{p}{d} \mathbb{Z} + jq_* \cdot \frac{\ell}{\gcd(\ell, N)} + \xi_* \right) \times \left\{ \frac{jdN}{p \gcd(\ell, N)} + \eta_* \right\}, \end{aligned}$$

where $d = \gcd(2, p)$, $q_* = \frac{(\frac{p}{d})^{\phi(\frac{2q}{d})-1}}{\frac{2q}{d}}$ and $(\xi_*, \eta_*) \in \mathbb{R} \times \mathbb{Z}_N$ satisfies $\xi_* a = \frac{m}{2} - \frac{\eta_*\ell}{N}$ for some odd integer m .

Proof Since $p \mid \frac{2N}{\gcd(\ell, N)}$, we have $\frac{p}{d} \mid \frac{N}{\gcd(\ell, N)}$. It follows that

$$\gcd\left(\frac{p}{d} \cdot \ell, N\right) = \frac{p}{d} \gcd(\ell, N).$$

By Lemma 3.5,

$$Z_2 \cap Z_3 = \bigcup_{j=0}^{\frac{p}{d} \cdot \gcd(\ell, N) - 1} 2 \left(\frac{p}{d} \mathbb{Z} + jq_* \cdot \frac{\ell}{\gcd(\ell, N)} \right) \times \left\{ \frac{jdN}{p \gcd(\ell, N)} \right\}.$$

Clearly,

$$\Lambda \subset (Z_2 \cap Z_3 + (\xi_*, \eta_*)) \cup (Z_2 \cap Z_3).$$

Since $Z_2 \cap Z_3$ is a subgroup of $\mathbb{Z} \times \mathbb{Z}_N$ and $(\pm \xi_*)a + \frac{(\pm \eta_*)\ell}{N} \in \mathbb{Z} + \frac{1}{2}$, by Lemma 3.5, we have

$$\Lambda - \Lambda \subset (Z_2 \cap Z_3 \pm (\xi_*, \eta_*)) \cup (Z_2 \cap Z_3) \subset Z_1 \cup (Z_2 \cap Z_3).$$

Since $\frac{p}{d} \mid \frac{N}{\gcd(\ell, N)}$ and $\gcd\left(\frac{N}{\gcd(\ell, N)}, \frac{\ell}{\gcd(\ell, N)}\right) = 1$, we have $\gcd\left(\frac{p}{d}, \frac{\ell}{\gcd(\ell, N)}\right) = 1$.

By the fact that $\gcd\left(\left(\frac{p}{d}\right)^{\phi\left(\frac{2|q|}{d}\right)} - 1, \frac{p}{d}\right) = 1$, we have $\gcd\left(q_*, \frac{p}{d}\right) = 1$. Therefore, the set

$$\left\{ jq_* \cdot \frac{\ell}{\gcd(\ell, N)} : 0 \leq j < \frac{p}{d} \right\}$$

forms a complete residue system modulo $\frac{p}{d}$. Consequently the elements in this set are pairwise distinct in the sense of modulo $\frac{p}{d}$. Clearly, $(\xi_*, \eta_*) \notin Z_2 \cap Z_3$. Hence, we have

$$\Lambda - \Lambda \subset (Z_1 \cup (Z_2 \cap Z_3)) \setminus Z_0.$$

And then the orthogonality of Λ follows from Lemma 3.2.

Now, we prove the completeness of Λ . For any $f \in L^2(\Omega)$, we define functions f_+ and f_- as follows:

$$f_+(x, y) = \frac{1}{2}(f(x, y) + f(x', y')), \quad f_-(x, y) = \frac{1}{2}(f(x, y) - f(x', y')), \quad (x, y) \in \Omega,$$

where

$$(x', y') = \begin{cases} (x + a, y + \ell), & (x, y) \in \left(0, \frac{1}{2}\right) \times \{0\}, \\ (x - a, y - \ell), & (x, y) \in \left(a, a + \frac{1}{2}\right) \times \{\ell\}. \end{cases}$$

Then

$$f(x, y) = f_+(x, y) + f_-(x, y), \quad (x, y) \in \Omega, \quad (3.15)$$

and

$$f_+(x, y) = f_+(x', y'), \quad f_-(x, y) = -f_-(x', y'), \quad (x, y) \in \Omega. \quad (3.16)$$

For $0 \leq j < \frac{p}{d}$, $k \in \mathbb{Z}$, denote

$$\lambda_{k,j} = (\xi_{k,j}, \eta_{k,j}) = \left(\frac{2p}{d}k + 2jq_* \cdot \frac{\ell}{\gcd(\ell, N)}, \frac{jdN}{p \gcd(\ell, N)} \right),$$

and

$$\lambda'_{k,j} = (\xi'_{k,j}, \eta'_{k,j}) = \left(\frac{2p}{d}k + 2jq_* \cdot \frac{\ell}{\gcd(\ell, N)} + \xi_*, \frac{jdN}{p \gcd(\ell, N)} + \eta_* \right).$$

Since $((0, \frac{1}{2}), 2\mathbb{Z})$ and $((0, \frac{1}{2}), 2\mathbb{Z} + \xi_*)$ are spectral pairs, we can establish the following results with a method similar to the proof of Proposition 3.4:

$$f_+(x, 0) = \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x, 0), \quad x \in \left(0, \frac{1}{2}\right),$$

and

$$f_-(x, 0) = \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} b_{\lambda'_{k,j}} e_{\lambda'_{k,j}}(x, 0), \quad x \in \left(0, \frac{1}{2}\right),$$

where

$$a_{\lambda_{k,j}} = 2 \int_0^{\frac{1}{2}} f_+(x, 0) e_{-\lambda_{k,j}}(x, 0) dx, \quad b_{\lambda'_{k,j}} = 2 \int_0^{\frac{1}{2}} f_-(x, 0) e_{-\lambda'_{k,j}}(x, 0) dx. \quad (3.17)$$

Since

$$\begin{aligned} \xi_{k,j}(-a) &\equiv \left(\frac{2p}{d}k + 2jq_* \cdot \frac{\ell}{\gcd(\ell, N)} \right) \cdot \left(-\frac{q}{p} \right) \\ &\equiv -\frac{2}{d}qk - j \cdot \left(\frac{p}{d} \right)^{\phi\left(\frac{2[q]\ell}{d}\right)-1} + j \cdot \frac{d\ell}{p \gcd(\ell, N)} \\ &\equiv j \cdot \frac{d\ell}{p \gcd(\ell, N)} \\ &\equiv \frac{\eta_{k,j}\ell}{N} \pmod{1}, \end{aligned}$$

it follows that

$$e_{\lambda_{k,j}}(x-a, 0) = e_{\lambda_{k,j}}(x, \ell), \quad x \in \left(a, a + \frac{1}{2}\right), \quad (3.18)$$

and

$$e_{\lambda'_{k,j}}(x-a, 0) = -e_{\lambda'_{k,j}}(x, \ell), \quad x \in \left(a, a + \frac{1}{2}\right). \quad (3.19)$$

Therefore, for $x \in \left(a, a + \frac{1}{2}\right)$,

$$f_+(x, \ell) = f_+(x - a, 0) = \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x - a, 0) = \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x, \ell),$$

and

$$f_-(x, \ell) = -f_-(x - a, 0) = -\sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} b_{\lambda'_{k,j}} e_{\lambda'_{k,j}}(x - a, 0) = \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} b_{\lambda'_{k,j}} e_{\lambda'_{k,j}}(x, \ell).$$

Hence,

$$f_+(x, y) = \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x, y), \quad (x, y) \in \Omega, \quad (3.20)$$

and

$$f_-(x, y) = \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} b_{\lambda'_{k,j}} e_{\lambda'_{k,j}}(x, y), \quad (x, y) \in \Omega. \quad (3.21)$$

Combining (3.20) and (3.21), by (3.15), we obtain:

$$f(x, y) = \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} a_{\lambda_{k,j}} e_{\lambda_{k,j}}(x, y) + \sum_{j=0}^{\frac{p}{d}-1} \sum_{k \in \mathbb{Z}} b_{\lambda'_{k,j}} e_{\lambda'_{k,j}}(x, y), \quad ; (x, y) \in \Omega.$$

□

3.2.3 $r = \frac{1}{2}$ and $\nu_2(\ell) < \nu_2(N)$

Proposition 3.7 Suppose Ω satisfies $r = \frac{1}{2}$, and $\nu_2(\ell) < \nu_2(N)$. Then Ω is a spectral set with spectrum

$$\Lambda = 2\mathbb{Z} \times \{0, \eta_0\},$$

where η_0 is a solution to Eq. (3.1).

Proof Note that if $\ell\eta \equiv \frac{N}{2} \pmod{N}$, then $\ell(-\eta) \equiv \frac{N}{2} \pmod{N}$. The orthogonality of Λ follows from Lemma 3.1.

We now verify the completeness of Λ . For any $f \in L^2(\Omega)$, we have $f(\cdot, 0) \in L^2\left(\left(0, \frac{1}{2}\right)\right)$ and $f(\cdot, l) \in L^2\left(\left(a, a + \frac{1}{2}\right)\right)$. Since $2\mathbb{Z}$ is the spectrum of $\left(0, \frac{1}{2}\right)$ and $\left(a, a + \frac{1}{2}\right)$, we have

$$f(x, 0) = \sum_{k \in \mathbb{Z}} a_k e^{2\pi i \cdot 2kx}, \quad x \in \left(0, \frac{1}{2}\right), \quad (3.22)$$

$$f(x, l) = \sum_{k \in \mathbb{Z}} b_k e^{2\pi i \cdot 2kx}, \quad x \in \left(a, a + \frac{1}{2}\right), \quad (3.23)$$

where $a_k = 2 \int_0^{\frac{1}{2}} f(x, 0) e^{-2\pi i \cdot 2kx} dx$ and $b_k = 2 \int_a^{a+\frac{1}{2}} f(x, l) e^{-2\pi i \cdot 2kx} dx$.

For $k \in \mathbb{Z}$, denote

$$\lambda_k = (2k, 0), \quad \text{and} \quad \lambda'_k = (2k, \eta_0).$$

Suppose c_k, d_k satisfy

$$\begin{cases} a_k = c_k + d_k, \\ b_k = c_k - d_k. \end{cases}$$

Then Eqs. (3.22) and (3.23) can be rewritten as

$$f(x, y) = \sum_{k \in \mathbb{Z}} c_k e_{\lambda_k}(x, y) + \sum_{k \in \mathbb{Z}} d_k e_{\lambda'_k}(x, y), \quad (x, y) \in \Omega.$$

□

Combining Propositions 3.4, 3.6, 3.7 and Theorem 1.2, we complete the proof of “tiling implies spectral”.

4 Spectral Implies Tiling

In this section, we provide a proof of “spectral implies tiling” when Ω is the union of two intervals in $\mathbb{R} \times \mathbb{Z}_N$.

4.1 Landau’s Density Condition for $\mathbb{R} \times \mathbb{Z}_N$

In the Euclidean setting, Landau’s necessary density conditions give sharp necessary conditions for sampling and interpolation in Paley–Wiener spaces. These results were extended to general locally compact abelian groups by Gröchenig et al. [16]. For our purposes, we need their theorem specialized to the group $\mathbb{R} \times \mathbb{Z}_N$, equipped with Haar measure ν . Besides its classical role in sampling and interpolation, this density theorem is also useful in the study of spectral sets, since it imposes necessary density constraints on possible spectra. In the proof of the “spectral implies tile” direction, this will be combined with the structure of the zero set $\mathcal{Z}_\Omega$ to show that Ω must satisfy one of the conditions in Theorem 1.2.

Let Ω be a measurable set in $\mathbb{R} \times \mathbb{Z}_N$ with $0 < \nu(\Omega) < \infty$ and let

$$\mathcal{B}(\Omega) = \{f \in L^2(\mathbb{R} \times \mathbb{Z}_N) : \text{supp}(\hat{f}) \subset \Omega\}.$$

A subset Λ of $\mathbb{R} \times \mathbb{Z}_N$ is called *uniformly discrete* if there exists an open set U such that the sets $\lambda + U$, $\lambda \in \Lambda$ are pairwise disjoint. A uniformly discrete set $\Lambda \subset \mathbb{R} \times \mathbb{Z}_N$

is called a *set of sampling* for $\mathcal{B}(\Omega)$ if there exists a constant $K > 0$ such that

$$\forall f \in \mathcal{B}(\Omega), \quad \|f\|^2 \leq K \sum_{\lambda \in \Lambda} |f(\lambda)|^2,$$

and Λ is called a *set of interpolation* for $\mathcal{B}(\Omega)$, if for every square-summable sequence $\{a_\lambda\}_{\lambda \in \Lambda}$, there exists an $f \in \mathcal{B}(\Omega)$ with $f(\lambda) = a_\lambda$, $\lambda \in \Lambda$.

Given two uniformly discrete sets Λ and Λ' and nonnegative numbers α and α' , we write $\alpha\Lambda \preceq \alpha'\Lambda'$ if for any positive number ε there exists a compact subset K of $\mathbb{R} \times \mathbb{Z}_N$ such that for every compact subset L we have

$$(1 - \varepsilon)\alpha \cdot \text{Card}(\Lambda \cap L) \leq \alpha' \cdot \text{Card}(\Lambda' \cap (K + L)).$$

The *lower uniform density* of Λ is defined as

$$\mathcal{D}^-(\Lambda) = \sup\{\alpha : \alpha(\mathbb{Z} \times \mathbb{Z}_N) \preceq \Lambda\},$$

and its *upper uniform density* is

$$\mathcal{D}^+(\Lambda) = \inf\{\alpha : \alpha\Lambda \preceq \mathbb{Z} \times \mathbb{Z}_N\}.$$

Theorem 4.1 ([16]) *Suppose Λ is a uniformly discrete subset of $\mathbb{R} \times \mathbb{Z}_N$ and Ω is a measurable set in $\mathbb{R} \times \mathbb{Z}_N$ with $0 < \nu(\Omega) < \infty$. Then*

- (1) *If Λ is a set of sampling for $\mathcal{B}(\Omega)$, then $\mathcal{D}^-(\Lambda) \geq N^{-1} \cdot \nu(\Omega)$.*
- (2) *If Λ is a set of interpolation for $\mathcal{B}(\Omega)$, then $\mathcal{D}^+(\Lambda) \leq N^{-1} \cdot \nu(\Omega)$.*

The appearance of the factor N^{-1} arises from the fact that the measure ν satisfies $\nu((0, 1) \times \{0\}) = 1$, while the Haar measure chosen in [16] satisfies $\nu((0, 1) \times \mathbb{Z}_N) = 1$.

For $r > 0$ and $x \in \mathbb{R}$, let $Q_r(x) = [-r/2, r/2] + x \times \mathbb{Z}_N$. The *lower Beurling density* of the uniformly discrete set Λ is defined as

$$\mathcal{D}_B^-(\Lambda) = \liminf_{r \rightarrow \infty} \inf_{x \in \mathbb{R}} \frac{\text{Card}(\Lambda \cap Q_r(x))}{r},$$

and its *upper Beurling density* is

$$\mathcal{D}_B^+(\Lambda) = \limsup_{r \rightarrow \infty} \sup_{x \in \mathbb{R}} \frac{\text{Card}(\Lambda \cap Q_r(x))}{r}.$$

If $\mathcal{D}_B^+(\Gamma) = \mathcal{D}_B^-(\Gamma)$, the common value $\mathcal{D}_B(\Gamma)$ is defined to be the *Beurling density*.

The following lemma provides a direct relationship between the two densities.

Lemma 4.2 *For every uniformly discrete set Λ , we have $\mathcal{D}^-(\Lambda) = N^{-1} \cdot \mathcal{D}_B^-(\Lambda)$ and $\mathcal{D}^+(\Lambda) = N^{-1} \cdot \mathcal{D}_B^+(\Lambda)$.*

Proof Let Λ be a uniformly discrete subset of $\mathbb{R} \times \mathbb{Z}_N$. Then, for every $\varepsilon > 0$, there exists a compact set $K = Q_R(0) = [-R/2, R/2] \times \mathbb{Z}_N$ such that

$$(1 - \varepsilon) \cdot \mathcal{D}^-(\Lambda) \cdot \text{Card}(\mathbb{Z} \times \mathbb{Z}_N \cap L) \leq \text{Card}(\Lambda \cap (L + Q_R(0)))$$

for every compact set L . Specializing to $L = Q_r(y)$, $y \in \mathbb{R}$, we get that

$$(1 - \varepsilon) \cdot \mathcal{D}^-(\Lambda) \cdot \inf_{y \in \mathbb{R}} \frac{\text{Card}(\mathbb{Z} \times \mathbb{Z}_N \cap Q_r(y))}{r + R} \leq \inf_{y \in \mathbb{R}} \frac{\text{Card}(\Lambda \cap Q_{r+R}(y))}{r + R}.$$

Taking the limit $r \rightarrow \infty$, we obtain $(1 - \varepsilon) \cdot N \cdot \mathcal{D}^-(\Lambda) \leq \mathcal{D}_B^-(\Lambda)$. So

$$N \cdot \mathcal{D}^-(\Lambda) \leq \mathcal{D}_B^-(\Lambda).$$

Conversely, for any given $\varepsilon > 0$, we may find $r_0 > 0$ such that

$$\frac{\text{Card}(\Lambda \cap Q_r(y))}{r} \geq (1 - \varepsilon) \cdot \mathcal{D}_B^-(\Lambda) \quad (4.1)$$

for every point $y \in \mathbb{R}$ and $r > r_0$. Now partition $R \times \mathbb{Z}_N$ into $Q_r(rk)$, $k \in \mathbb{Z}$, whose interiors are disjoint. Given a compact set $L \subset \mathbb{R} \times \mathbb{Z}_N$, there exists finitely many $k_j \in \mathbb{Z}$, $j = 1, \dots, J$, such that

$$L \subset \bigcup_{j=1}^J Q_r(rk_j) \subset L + Q_{2r}(0).$$

Then by (4.1), we have

$$\text{Card}(\Lambda \cap (L + Q_{2r}(0))) \geq \sum_{j=1}^J \text{Card}(\Lambda \cap Q_r(rk_j)) \geq (1 - \varepsilon)rJ \cdot \mathcal{D}_B^-(\Lambda).$$

Since $(r + 1)N \geq \text{Card}(\mathbb{Z} \times \mathbb{Z}_N \cap Q_r(rk_j))$, it follows that

$$\text{Card}(\Lambda \cap (L + Q_{2r}(0))) \geq (1 - \varepsilon)(1 + 1/r)^{-1}N^{-1} \cdot \mathcal{D}_B^-(\Lambda) \cdot \text{Card}(\mathbb{Z} \times \mathbb{Z}_N \cap L).$$

We may choose ε arbitrarily small enough and r arbitrarily large, and hence

$$\mathcal{D}^-(\Lambda) \geq N^{-1} \cdot \mathcal{D}_B^-(\Lambda).$$

The identity $\mathcal{D}^+(\Lambda) = N^{-1} \cdot \mathcal{D}_B^+(\Lambda)$ is proved similarly. $\square$

Combining Theorem 4.1 and Lemma 4.2, we obtain the spectrum density condition.

Corollary 4.3 Suppose Ω is a spectral set and Λ is the spectrum. Then $\mathcal{D}_B(\Lambda) = \nu(\Omega)$.

4.2 Classification of Spectrum

We now begin the proof of “spectral implies tiling”. Suppose Ω satisfies (1.1), and $\Lambda = \{\lambda_k = (\xi_k, \eta_k) : k \in \mathbb{Z}\}$ is a spectrum for Ω with $\lambda_0 = (0, 0)$. Recall that

$$\Omega = (0, r) \times \{0\} \cup (a, a + 1 - r) \times \{\ell\}, \quad \frac{1}{2} \leq r < 1, \quad 1 \leq \ell \leq N - 1.$$

Since $\nu(\Omega) = 1$, by Corollary 4.3, we have $\mathcal{D}_B(\Lambda) = 1$.

Recall that $Z_0 = \{(0, j) : 0 < j < N\}$ and

$$\mathcal{Z}_\Omega = \{(0, 0)\} \cup \{(\xi, \eta) \in \mathbb{R} \times \mathbb{Z}_N : \widehat{1_\Omega}(\xi, \eta) = 0\}.$$

We discuss into two cases: $\mathcal{Z}_\Omega \cap Z_0 = \emptyset$ and $\mathcal{Z}_\Omega \cap Z_0 \neq \emptyset$.

By Lemma 3.1, we know if $\mathcal{Z}_\Omega \cap Z_0 \neq \emptyset$, then $r = \frac{1}{2}$ and $\nu_2(\ell) < \nu_2(N)$. Now Ω is a tile by Theorem 1.2.

From now on, we assume

$$\mathcal{Z}_\Omega \cap Z_0 = \emptyset.$$

Since $(0, 0) \in \Lambda$, we have $\Lambda \subset \Lambda - \Lambda \subset \mathcal{Z}_\Omega$. Then by Lemma 3.2,

$$\Lambda \subset \Lambda - \Lambda \subset (Z_1 \cup Z_2 \cup Z_3) \setminus Z_0. \quad (4.2)$$

Denote $\lambda_{kk'} = \lambda_k - \lambda_{k'}$, $\Lambda - \Lambda = \{\lambda_{kk'} : k, k' \in \mathbb{Z}\}$. Similarly, we denote $\xi_{kk'} = \xi_k - \xi_{k'}$ and $\eta_{kk'} = \eta_k - \eta_{k'}$.

The following lemma follows from the orthogonality of Λ .

Lemma 4.4 *At least one of the following holds:*

$$\Lambda \subset (Z_1 \cup Z_2) \setminus Z_0,$$

$$\Lambda \subset (Z_1 \cup Z_3) \setminus Z_0.$$

Proof If $Z_2 \subset Z_3$, by (4.2) we have $\Lambda \subset (Z_1 \cup Z_3) \setminus Z_0$. Suppose there exists a $\lambda_{k_1} \in (Z_2 \setminus Z_3) \cap \Lambda$. We will show that for any $\lambda_{k_2} \in Z_3 \cap \Lambda$, it holds that $\lambda_{k_2} \in Z_1$ or $\lambda_{k_2} \in Z_2$.

Let $\lambda_{k_2} \in Z_3 \cap \Lambda$. Then, by orthogonality, $\lambda_{k_1 k_2} = \lambda_{k_1} - \lambda_{k_2} \in \mathcal{Z}_\Omega$. By (4.2), $\lambda_{k_1 k_2} \in (Z_1 \cup Z_2 \cup Z_3) \setminus Z_0$. Thus we need to consider the following three cases:

- (1) If $\lambda_{k_1 k_2} \in Z_2$, then $\lambda_{k_2} = \lambda_{k_1} - \lambda_{k_1 k_2} \in Z_2$.
- (2) If $\lambda_{k_1 k_2} \in Z_3$, then $\lambda_{k_1} = \lambda_{k_2} + \lambda_{k_1 k_2} \in Z_3$, which contradicts $\lambda_{k_1} \notin Z_3$.
- (3) If $\lambda_{k_1 k_2} \in Z_1$, then $\xi_{k_1 k_2}(1 - 2r) \in \mathbb{Z}$. Since $\xi_{k_1 k_2} = \xi_{k_1} - \xi_{k_2} \in \mathbb{Z}$, and $\xi_{k_1} r \in \mathbb{Z}$, it follows that

$$2\xi_{k_2} r = 2\xi_{k_1} r + \xi_{k_1 k_2}(1 - 2r) - \xi_{k_1 k_2} \in \mathbb{Z}.$$

If $\xi_{k_2} r \in \mathbb{Z}$, then, since $\xi_{k_2} \in \mathbb{Z}$, it follows that $\lambda_{k_2} \in Z_2$. On the other hand, if $\xi_{k_2} r \in \mathbb{Z} + \frac{1}{2}$, then since $\xi_{k_2}(a - r) \in \mathbb{Z} - \frac{\eta_{k_2} \ell}{N}$, we have

$$\xi_{k_2} a = \xi_{k_2} r + \xi_{k_2}(a - r) \in \mathbb{Z} + \frac{1}{2} - \frac{\eta_{k_2} \ell}{N}.$$

Because $\xi_{k_2} \in \mathbb{Z}$ and $\xi_{k_2}(1 - 2r) = \xi_{k_2} - 2\xi_{k_2}r \in \mathbb{Z}$, it follows that $\lambda_{k_2} \in Z_1$.

□

The following lemma rules out the possibility of Λ being a subset of $Z_2 \setminus Z_0$, which is a direct consequence of the spectrum density condition.

Lemma 4.5 $\Lambda \subset Z_2 \setminus Z_0$ is not possible.

Proof Let $r = \frac{q_r}{p_r}$ with $(p_r, q_r) = 1$. By Lemma 3.2, we have

$$Z_2 = \bigcup_{j=0}^{N-1} \{\xi \in \mathbb{Z} : \xi r \in \mathbb{Z}\} \times \{j\} = (r^{-1}\mathbb{Z} \cap \mathbb{Z}) \times \mathbb{Z}_N = p_r\mathbb{Z} \times \mathbb{Z}_N.$$

Now, suppose $\Lambda \subset Z_2 \setminus Z_0$. By (4.2), we have $(\Lambda - \Lambda) \cap Z_0 = \emptyset$. Since Z_2 is a subgroup of $\mathbb{Z} \times \mathbb{Z}_N$, we have $\Lambda - \Lambda \subset Z_2 \setminus Z_0$, which implies the first components to two point in Λ are distinct. Hence, the Beurling density of Λ is at most $1/p_r$. Since $\frac{1}{2} \leq r < 1$, it follows that $1/p_r < 1$. This contradicts $\mathcal{D}_B(\Lambda) = 1$. □

The following lemma provides a necessary condition for $a - r$ when $\Lambda \subset Z_3 \setminus Z_0$.

Lemma 4.6 If $\Lambda \subset Z_3 \setminus Z_0$, then $a - r \in \frac{\gcd(\ell, N)}{N}\mathbb{Z}$.

Proof Clearly, $a - r \in \mathbb{Q}$. Otherwise $Z_3 = \left\{ \left(0, \frac{jN}{\gcd(\ell, N)} \right) : 0 \leq j \leq \gcd(\ell, N) - 1 \right\}$ would only have a finite number of points, which is a contradiction.

Now we assume $a - r \in \mathbb{Q} \setminus \frac{\gcd(\ell, N)}{N}\mathbb{Z}$. Let $a - r = \frac{q}{p}$ with $\gcd(p, q) = 1$, $p > 0$. Then $p \nmid \frac{N}{\gcd(\ell, N)}$ and $p \neq 1$. By Lemma 3.3

$$Z_3 = \bigcup_{j=0}^{\gcd(p\ell, N)-1} \left(p\mathbb{Z} + jq_* \cdot \frac{p\ell}{\gcd(p\ell, N)} \right) \times \left\{ \frac{jN}{\gcd(p\ell, N)} \right\}, \quad (4.3)$$

where $q_* = \frac{p^{\phi(q)} - 1}{q} \in \mathbb{Z} \setminus \{0\}$.

By (4.2) and the fact that Z_3 is a subgroup of $\mathbb{Z} \times \mathbb{Z}_N$, we know that

$$\Lambda - \Lambda \subset Z_3 \setminus Z_0.$$

Therefore, $\xi_{k_1} \neq \xi_{k_2}$ for any $\lambda_{k_1}, \lambda_{k_2} \in \Lambda$. Then we consider the congruence equation

$$\left(q_* \cdot \frac{p\ell}{\gcd(p\ell, N)} \right) \eta \equiv 0 \pmod{p},$$

or equivalently,

$$(q_*\ell)\eta \equiv 0 \pmod{\gcd(p\ell, N)}. \quad (4.4)$$

Since $\gcd(q_*, p) = 1$, it follows that

$$\gcd(q_*\ell, \gcd(p\ell, N)) = \gcd(q_*\ell, p\ell, N) = \gcd(\gcd(q_*\ell, p\ell), N) = \gcd(\ell, N).$$

Therefore, the solution to the congruence equation (4.4) is given by

$$\eta_t \equiv \frac{\gcd(p\ell, N)}{\gcd(\ell, N)} t \pmod{\gcd(p\ell, N)}, \quad 0 \leq t < \gcd(p\ell, N). \quad (4.5)$$

Then (4.3) can be rewritten as

$$Z_3 = \bigcup_{t=0}^{\gcd(p\ell, N)} \bigcup_{j=0}^{\frac{\gcd(p\ell, N)}{\gcd(\ell, N)} - 1} \left(p\mathbb{Z} + jq_* \cdot \frac{p\ell}{\gcd(p\ell, N)} \right) \times \left\{ (j + \eta_t) \cdot \frac{N}{\gcd(p\ell, N)} \right\}. \quad (4.6)$$

When the first components are distinct, by (4.6), the Beurling density of Λ is at most

$$\frac{\gcd(p\ell, N)}{p \gcd(\ell, N)},$$

which is strictly less than 1 since $p \nmid \frac{N}{\gcd(\ell, N)}$, a contradiction to $\mathcal{D}_B(\Lambda) = 1$.

Therefore, we have

$$a - r \in \frac{\gcd(\ell, N)}{N} \mathbb{Z}.$$

□

The following corollary is a direct consequence of Theorem 1.2 and Lemma 4.6.

Corollary 4.7 *Suppose $\Lambda \subset Z_3 \setminus Z_0$. Then Ω is a tile.*

From now on, we assume that:

$$\Lambda \not\subset Z_2 \setminus Z_0, \quad \Lambda \not\subset Z_3 \setminus Z_0.$$

Under this assumption, the following lemma characterizes the structure of Λ .

Lemma 4.8 *If $\Lambda \not\subset Z_2 \setminus Z_0$ and $\Lambda \not\subset Z_3 \setminus Z_0$, then*

$$\Lambda \subset \Lambda - \Lambda \subset (Z_1 \cup (Z_2 \cap Z_3)) \setminus Z_0.$$

Proof To prove $\Lambda \subset (Z_1 \cup (Z_2 \cap Z_3)) \setminus Z_0$, according to Lemma 4.4, it suffices to prove that:

- (1) If $\Lambda \cap (Z_1 \setminus Z_2) \neq \emptyset$, then $\Lambda \cap Z_2 \subset \Lambda \cap Z_3$;
- (2) If $\Lambda \cap (Z_1 \setminus Z_3) \neq \emptyset$, then $\Lambda \cap Z_3 \subset \Lambda \cap Z_2$.

We will prove (1), the proof of (2) is almost the same.

Assume $\lambda_{k_1} \in (Z_1 \setminus Z_2) \cap \Lambda$. For any $\lambda_{k_2} \in Z_2 \cap \Lambda$, we need to show $\lambda_{k_2} \in Z_3$. By Lemma 3.2, $\lambda_{k_1 k_2} \in (Z_1 \cup Z_2 \cup Z_3) \setminus Z_0$. We now consider the following three cases:

- (i) If $\lambda_{k_1 k_2} \in Z_2$, then $\lambda_{k_1} = \lambda_{k_1 k_2} + \lambda_{k_2} \in Z_2$. This contradicts the assumption that $\lambda_{k_1} \notin Z_2$.
- (ii) If $\lambda_{k_1 k_2} \in Z_1$, then $\xi_{k_1 k_2} a \in \mathbb{Z} + \frac{1}{2} - \frac{\eta_{k_1 k_2} \ell}{N}$. Since $\xi_{k_1} a \in \mathbb{Z} + \frac{1}{2} - \frac{\eta_{k_1} \ell}{N}$, it follows that $\xi_{k_2} a \in \mathbb{Z} - \frac{\eta_{k_2} \ell}{N}$. Since $\xi_{k_2} r \in \mathbb{Z}$, then $\xi_{k_2} (a - r) \in \mathbb{Z} - \frac{\eta_{k_2} \ell}{N}$. Moreover, since $\xi_{k_2} \in \mathbb{Z}$, we obtain that $\lambda_{k_2} \in Z_3$.
- (iii) If $\lambda_{k_1 k_2} \in Z_3$, from $\xi_{k_1} = \xi_{k_1 k_2} + \xi_{k_2} \in \mathbb{Z}$ we deduce that

$$2\xi_{k_1} r = -\xi_{k_1} (1 - 2r) + \xi_{k_1} \in \mathbb{Z}.$$

Then $\xi_{k_1} r \in \mathbb{Z} \cup \left(\mathbb{Z} + \frac{1}{2}\right)$. If $\xi_{k_1} r \in \mathbb{Z}$, then $\lambda_{k_1} \in Z_2$, which contradicts $\lambda_{k_1} \notin Z_2$. Therefore, $\xi_{k_1} r \in \mathbb{Z} + \frac{1}{2}$. Since $\lambda_{k_1} \in Z_1$, we have $\xi_{k_1} a \in \mathbb{Z} + \frac{1}{2} - \frac{\eta_{k_1} \ell}{N}$ and then

$$\xi_{k_1} (a - r) = \xi_{k_1} a - \xi_{k_1} r \in \mathbb{Z} - \frac{\eta_{k_1} \ell}{N}.$$

Hence $\lambda_{k_1} \in Z_3$ and $\lambda_{k_2} = \lambda_{k_1} - \lambda_{k_1 k_2} \in Z_3$.

For $k \in \mathbb{Z}$, define $\Lambda_k = \Lambda - \lambda_k = \{\lambda_{jk} : j \in \mathbb{Z}\}$. Then Λ_k is also a spectrum of Ω , and $(0, 0) \in \Lambda_k$. Therefore, the conclusions about Λ hold true for Λ_k as well. That is,

$$\Lambda - \Lambda = \bigcup_{k \in \mathbb{Z}} \Lambda_k \subset (Z_1 \cup (Z_2 \cap Z_3)) \setminus Z_0.$$

□

The following lemma gives the necessary conditions on r and a required for $\mathcal{D}_B(\Lambda) = 1$.

Lemma 4.9 *If $\Lambda \not\subset Z_2 \setminus Z_0$ and $\Lambda \not\subset Z_3 \setminus Z_0$, then $r = \frac{1}{2}$, $a \in \mathbb{Q} \setminus \{0\}$.*

Proof If $a = 0$, then $Z_1 = \emptyset$, a contradiction.

If $a \notin \mathbb{Q}$ or $r \notin \mathbb{Q}$, then

$$Z_2 \cap Z_3 = \left\{ \left(0, \frac{jN}{\gcd(\ell, N)} \right) : 0 \leq j \leq \gcd(\ell, N) - 1 \right\}.$$

For any $\lambda_{k_1}, \lambda_{k_2} \in \Lambda \cap Z_1$, we have $\xi_{k_i} a \in \mathbb{Z} + \frac{1}{2} - \frac{\eta_{k_i} \ell}{N}$, $i = 1, 2$. Then $\xi_{k_1 k_2} a \in \mathbb{Z} - \frac{\eta_{k_1 k_2} \ell}{N}$, which implies $\lambda_{k_1 k_2} \notin Z_1$. By Lemma 4.8, $\lambda_{k_1 k_2} \in Z_2 \cap Z_3$, thus $\xi_{k_1 k_2} = 0$, and

$$\eta_{k_1 k_2} \in \left\{ \frac{jN}{\gcd(\ell, N)} : 0 \leq j \leq \gcd(\ell, N) - 1 \right\}.$$

Consequently,

$$\Lambda \cap Z_1 \subset \left\{ \left(\xi_*, \frac{jN}{\gcd(\ell, N)} + \eta_* \right) : 0 \leq j \leq \gcd(\ell, N) - 1 \right\},$$

for some $(\xi_*, \eta_*) \in \mathbb{R} \times \mathbb{Z}_N$. Hence, $\Lambda \cap (Z_1 \cup (Z_2 \cap Z_3))$ contains only finitely many points, which contradicts $\mathcal{D}_B(\Lambda) = 1$.

Let $r = \frac{q_r}{p_r}$ with $(p_r, q_r) = 1$. Since $\frac{1}{2} \leq r < 1$, it follows that $p_r \geq 2$. Note that

$$Z_2 \cap Z_3 \subset Z_2 = p_r \mathbb{Z} \times \mathbb{Z}_N.$$

Similarly, we obtain that $\Lambda \cap Z_1 \subset (p_r \mathbb{Z} + \xi_*) \times \mathbb{Z}_N$. Therefore, by Lemma 4.4, we have

$$\Lambda \subset \Lambda - \Lambda \subset (((p_r \mathbb{Z} + \xi_*) \cup p_r \mathbb{Z}) \times \mathbb{Z}_N) \setminus Z_0. \quad (4.7)$$

Finally, by $\mathcal{D}_B(\Lambda) = 1$ and (4.7), we conclude that $p_r = 2$ and hence $r = \frac{1}{2}$. $\square$

The following lemma follows from (4.9), Lemma 4.8 and the orthogonality of Λ . It further refines the structure of Λ under the given assumptions.

Lemma 4.10 *If $\Lambda \not\subset Z_2 \setminus Z_0$, $\Lambda \not\subset Z_3 \setminus Z_0$, then*

$$\Lambda \cap Z_1 \subset Z_2 \cap Z_3 + (\xi_*, \eta_*) \quad (4.8)$$

for some $\xi_* \in \mathbb{R}$ and $\eta_* \in \mathbb{Z}_N$.

Proof By Lemma 4.9, we have $a \in \mathbb{Q} \setminus \{0\}$ and $r = \frac{1}{2}$. Write

$$a = \frac{q}{p}, \quad \text{with } p > 0, \quad \gcd(p, q) = 1.$$

By Lemma 3.5, we have

$$Z_2 \cap Z_3 = \bigcup_{j=0}^{\gcd\left(\frac{p}{d} \cdot \ell, N\right)-1} 2 \left(\frac{p}{d} \mathbb{Z} + jq_* \cdot \frac{\frac{p}{d} \cdot \ell}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \right) \times \left\{ \frac{jN}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \right\}, \quad (4.9)$$

where $d = (p, 2)$ and $q_* = \frac{(\frac{p}{d})^{\phi\left(\frac{2q}{d}\right)-1}}{\frac{2q}{d}} \in \mathbb{Z}$.

For any $\lambda_{k_1}, \lambda_{k_2} \in \Lambda \cap Z_1$, by the definition of Z_1 , we have

$$\xi_{k_i} a \in \mathbb{Z} + \frac{1}{2} - \frac{\eta_{k_i} \ell}{N}, \quad i = 1, 2.$$

Consequently, $\xi_{k_1 k_2} a \in \mathbb{Z} - \frac{\eta_{k_1 k_2} \ell}{N}$, which implies $\lambda_{k_1 k_2} \notin Z_1$. By Lemma 4.8, we have $\lambda_{k_1 k_2} \in Z_2 \cap Z_3$. By (4.9), we have

$$\eta_{k_1 k_2} \equiv 0 \left(\text{mod } \frac{N}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \right),$$

and then there exists $0 \leq j_{k_1}, j_{k_2} < \gcd\left(\frac{p}{d} \cdot \ell, N\right)$ and $\eta_* \in \mathbb{Z}_N$ such that

$$\eta_{k_1} = \frac{j_{k_1} N}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} + \eta_*, \quad \eta_{k_2} = \frac{j_{k_2} N}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} + \eta_*.$$

By (4.9) again,

$$\xi_{k_1 k_2} \equiv 2j_{k_1 k_2} q_* \cdot \frac{\frac{p}{d} \cdot \ell}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \left(\text{mod } \frac{2p}{d} \right),$$

where $j_{k_1 k_2} = j_{k_1} - j_{k_2}$. This implies that

$$\xi_{k_1} - 2j_{k_1} q_* \cdot \frac{\frac{p}{d} \cdot \ell}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \equiv \xi_{k_2} - 2j_{k_2} q_* \cdot \frac{\frac{p}{d} \cdot \ell}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \left(\text{mod } \frac{2p}{d} \right).$$

Thus,

$$\Lambda \cap Z_1 \subset \bigcup_{j=0}^{\gcd\left(\frac{p}{d} \cdot \ell, N\right)-1} \left(2 \left(\frac{p}{d} \mathbb{Z} + j q_* \cdot \frac{\frac{p}{d} \cdot \ell}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} \right) + \xi_* \right) \times \left\{ \frac{j N}{\gcd\left(\frac{p}{d} \cdot \ell, N\right)} + \eta_* \right\}$$

for some $\xi_* \in \mathbb{R}$ and $\eta_* \in \mathbb{Z}_N$. $\square$

The following lemma summarizes the previous results and provides the condition that a must satisfy when Ω is a spectral set.

Lemma 4.11 *If $\Lambda \not\subset Z_2 \setminus Z_0$, $\Lambda \not\subset Z_3 \setminus Z_0$, then*

$$r = \frac{1}{2} \quad \text{and} \quad a \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z} \setminus \{0\}.$$

Proof By Lemmas 4.8, 4.10 and (4.9), we have

$$\Lambda \subset (Z_2 \cap Z_3) \cup (Z_2 \cap Z_3 + (\xi_*, \eta_*))$$

and $(\Lambda - \Lambda) \cap Z_0 = \emptyset$, which implies for any $\lambda_{k_1}, \lambda_{k_2} \in \Lambda$, $\xi_{k_1} \neq \xi_{k_2}$.

Similar to the proof of Lemma 4.6, we obtain that

$$a \in \frac{\gcd(\ell, N)}{2N} \mathbb{Z} \setminus \{0\}.$$

□

The following corollary is a direct consequence of Theorem 1.2 and Lemma 4.11.

Corollary 4.12 *Suppose $\Lambda \not\subset Z_2 \setminus Z_0$ and $\Lambda \not\subset Z_3 \setminus Z_0$. Then Ω is a tile.*

“Spectral implies tiling” follows from Theorem 1.2, Corollaries 4.7, and 4.12.

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Declarations

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